

Project Initialization and Planning Phase

Date	02 JULY 2026
Team ID	CCAP-01
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to automate the credit card approval decision-making process using machine learning, reducing manual review effort and improving consistency and accuracy. It addresses the inefficiencies of manual application screening, promising faster turnaround, fewer errors, and improved compliance. Key features include a multi-algorithm credit classification model and a real-time, cloud-deployed prediction interface.

Project Overview	
Objective	The primary objective is to automate and improve the credit card approval process by applying machine learning techniques, ensuring faster, more consistent, and more accurate approval/rejection decisions.
Scope	The project covers data collection, feature engineering, model training and comparison across multiple classification algorithms, and deployment of the best model as a real-time, cloud-hosted web application for analysts, compliance officers, and applicants.

Problem Statement	
Description	Manual review of the high daily volume of credit card applications is time-consuming and prone to human error, leading to inconsistent decisions and delayed turnaround for applicants.
Impact	Automating the decision process reduces processing time and error rates, improves regulatory compliance through consistent risk classification, and increases customer satisfaction by providing instant eligibility feedback.

Proposed Solution	
Approach	Train and compare multiple machine learning classification models on historical applicant data to predict credit card approval outcomes, then deploy the best-performing model as a real-time prediction service.
Key Features	- Implementation of multiple ML credit classification models: Logistic Regression, RandomForest, XGBoost (Gradient Boosting), and Decision Tree. - Real-time approval/rejection prediction through a Flask web application. - Cloud deployment via IBM Watson Machine Learning for scalable, real-time predictions. - Feature engineering pipeline converting multi-class payment status codes into binary risk labels for high-risk applicant identification.

Resource Requirements

Hardware

Resource Type	Description	Specification/Allocation
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	16 GB
Storage	Disk space for data, models, and logs	1 TB SSD

Software

Frameworks	Flask
Libraries	scikit-learn, pandas, numpy, xgboost, matplotlib, seaborn
Development Environment	Jupyter Notebook, PyCharm

Data

Data Source, size, format	Kaggle - Credit Card Approval Prediction dataset, 438,557 records, csv
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